



Quantifying the Court: Modeling cross-court groundstroke accuracy using weighted target zones and machine learning in men's collegiate tennis



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Abstract

This project explores the relationship between ball characteristics, fatigue, and how they relate to the accuracy of a crosscourt groundstroke. Data was collected using SwingVision software and manually documenting where the ball landed in the court on previously determined regions. This study introduced a reproducible weighted target-zone scoring system and analytical workflow for measuring tennis groundstroke accuracy. While limitations in the experimental design prevented definitive conclusions about the effects of stroke type and fatigue, the framework establishes a strong foundation for future studies.

Introduction

Limited coaching resources force Willamette tennis players to rely on subjective intuition rather than objective performance data to adjust their match strategies. By identifying the variables with the greatest influence on shot accuracy, this project provides collegiate players and coaches with data-driven insights to optimize training and improve real-time tactical adjustments based on stroke speed and accuracy trade-offs.

The primary internal stakeholders are the Willamette coaching staff and players, who directly benefit from these objective metrics to make informed match decisions despite limited on-court resources. External stakeholders include independent tennis players who can utilize this tracking experiment for self-evaluation.

Background

Prior tennis analytics heavily favor elite, high-resource environments. Kovalchik and Reid (2018) used professional tracking data to prove high stroke speeds dictate point success, while Lei et al. (2024) leveraged advanced machine learning to forecast professional match outcomes. In sports science, Fiorilli et al. (2020) tracked tennis fatigue but isolated their metrics to off-court fitness drills rather than active, technical play.

This project bridges the gap by offering a low-resource, self-administered framework. Instead of relying on expensive tracking infrastructure, this study applies a Random Forest classifier to model how continuous fatigue and stroke velocity interact to affect on-court accuracy for collegiate athletes.

Methods and Evaluation

Approach:

- Compared six regression models (Linear Regression, KNN, SVR, Random Forest, Gradient Boosting, XGBoost) rather than committing to one in advance
- Predictors: stroke type, fatigue, UTR, handedness; target: continuous accuracy score
- Benchmarked against a naive baseline (predicts sample mean) since collegiate-level proficiency can make simple predictions look deceptively reasonable without one

Validation:

- Leave-one-session-out cross-validation (each fold withholds one player's full session, not a random subset of shots)
- Necessary since shots from the same player aren't independent; a random split would overstate performance
- Evaluated on MAE, RMSE, and R^2 vs. baseline; hyperparameters tuned on the same session-held-out folds

Findings:

- Best model: Random Forest, $R^2 = -0.190$, MAE = 0.189 (tuned)
- SVR: $R^2 = -0.196$; Gradient Boosting: $R^2 = -0.214$; baseline: $R^2 = -0.304$
- No model reached positive R^2 (none reliably predicted a held-out player), but 3 of 5 beat baseline
- Winning Random Forest used a shallow tree (max depth = 3), favoring simplicity given limited signal
- Feature importance (tuned models): UTR 66-72%, fatigue x stroke interaction 12-14%, fatigue alone 11-18%, handedness under 5%, stroke type ~0%

Data, Engineering, and Ethics

Sources & scale: Data was self collected and collected using Swing Vision, approximately 1000 shots (around 200 shots per participant). There were five participants, all were men who are part of the Willamette community.

Ethics & responsible use: Consent forms were required to be filled out before data collection could start, personal identifiers were anonymized, biased candidate pool (needed mostly experienced players), physical risks were involved but acknowledged in consent form, data is password protected / recorded sessions will be deleted after project is completed / participants can access there session recording if requested.

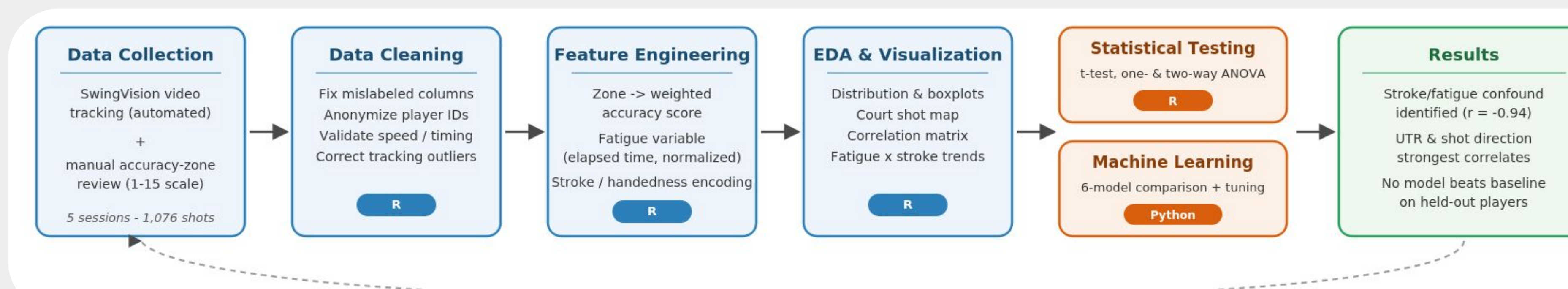


Figure 1. (Above) Pipeline sketch outlining data collection, cleaning, analytical methods, and results.

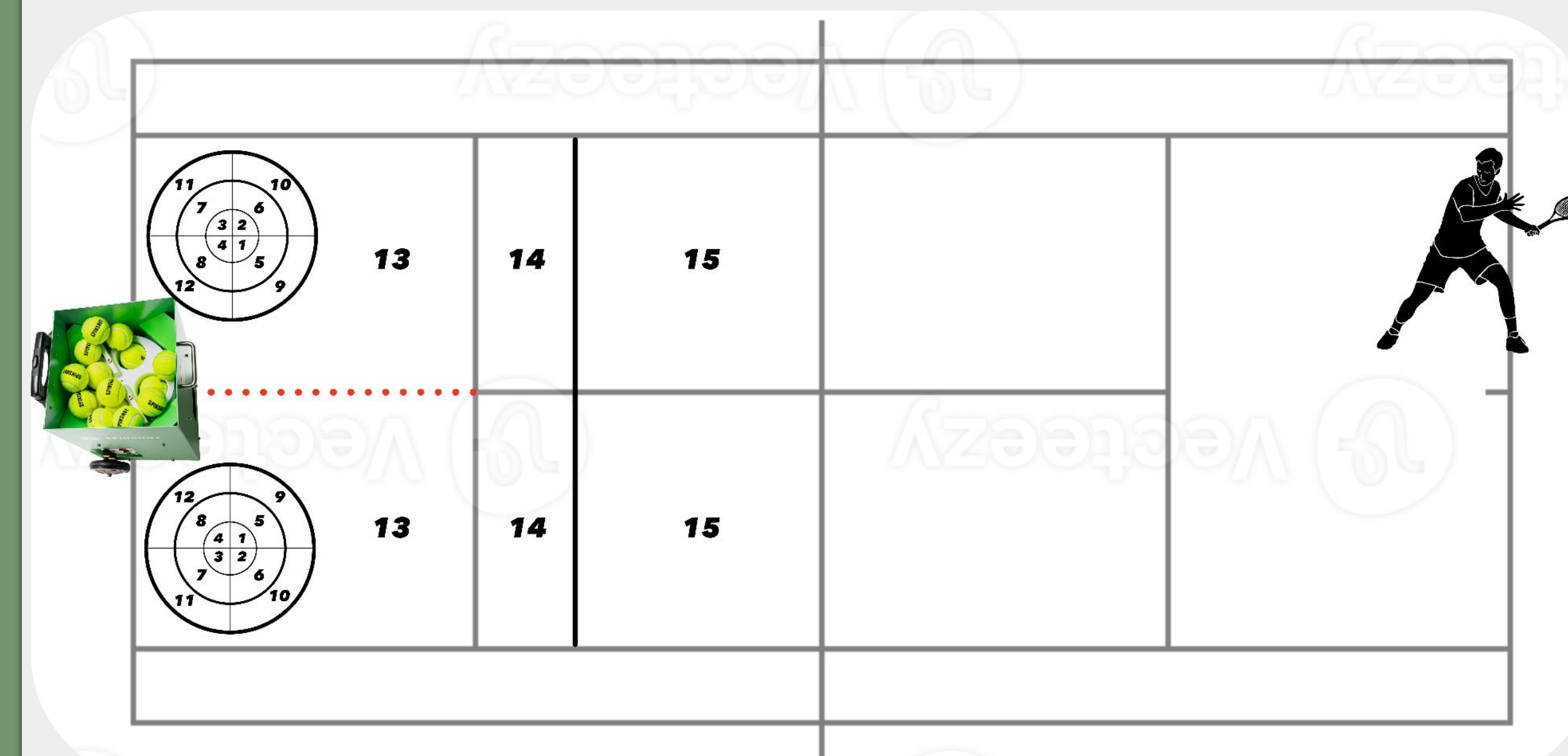
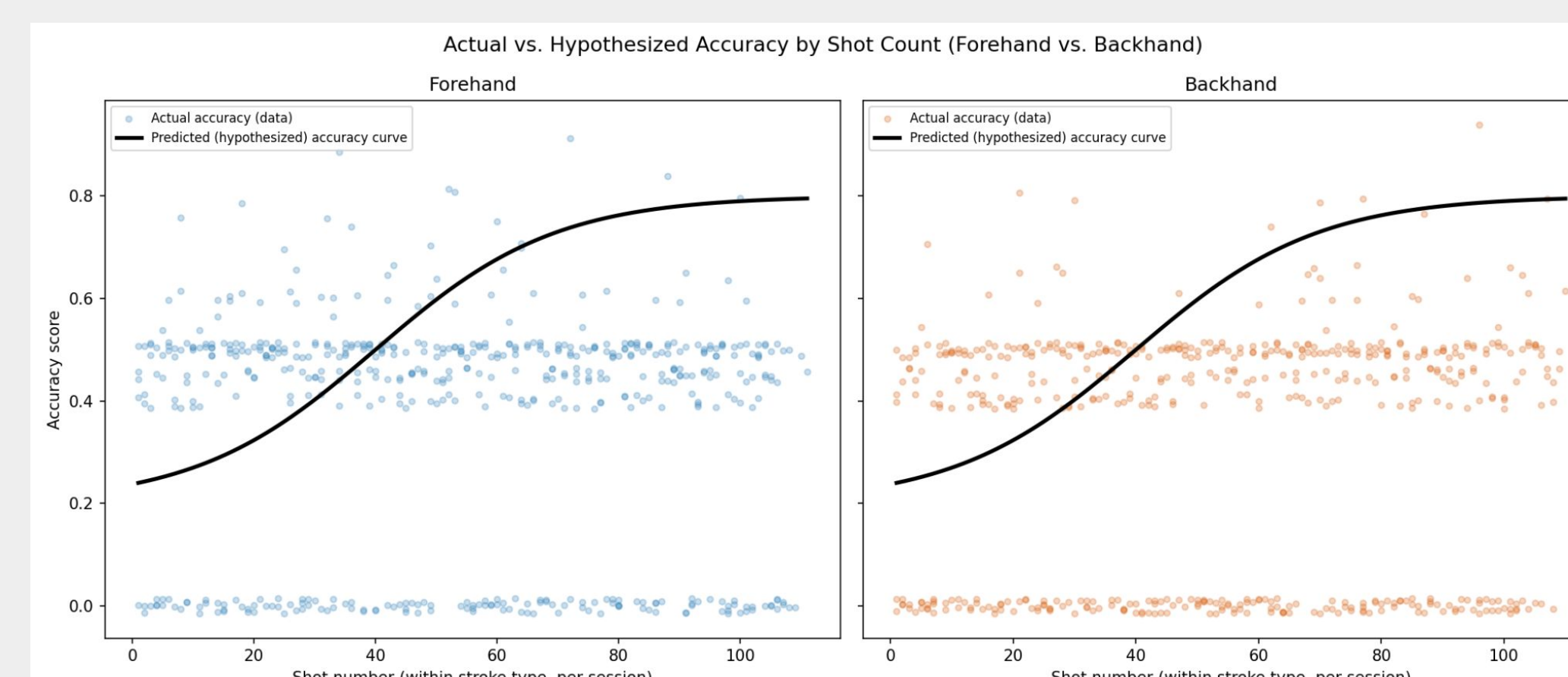


Figure 2. (Above) Court diagram defining accuracy zones for both forehand and backhand strokes.

Figure 3. (Right) Accuracy weighting per zone.

0	0.00
1	0.95
2	0.90
3	0.90
4	0.85
5	0.80
6	0.75
7	0.75
8	0.70
9	0.65
10	0.60
11	0.60
12	0.55
13	0.50
14	0.45
15	0.40

Figure 6. (Below) Actual accuracy scores by shot number within each stroke type (forehand, backhand), plotted against a hypothesized (non-fitted) sigmoid curve predicting a rise from 0.2 to 0.8 over the first ~65 shots.



Results and Visualizations

Swing Vision Variables: Shot, Stroke, Spin, Speed (MPH), Bounce Depth, Bounce Zone, Bounce Side, Bounce (x), Bounce (y), Hit Depth, Hit Zone, Hit Side, Hit (x), Hit (y), Hit (z), Result, Start Time, Video Time

Self Determined Variables: Session ID, Player ID, UTR, Handedness, Accuracy Zone

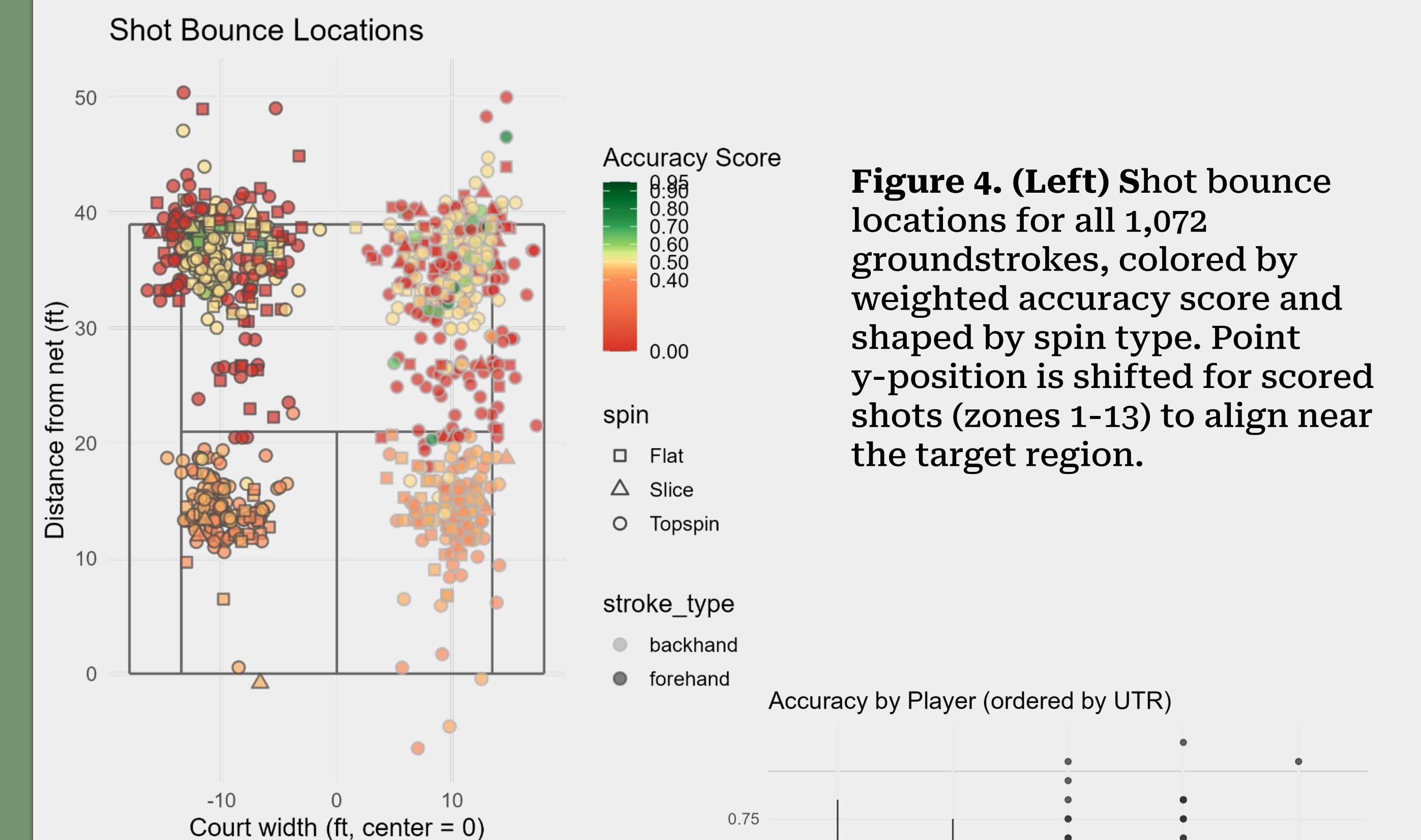


Figure 4. (Left) Shot bounce locations for all 1,072 groundstrokes, colored by weighted accuracy score and shaped by spin type. Point y-position is shifted for scored shots (zones 1-13) to align near the target region.

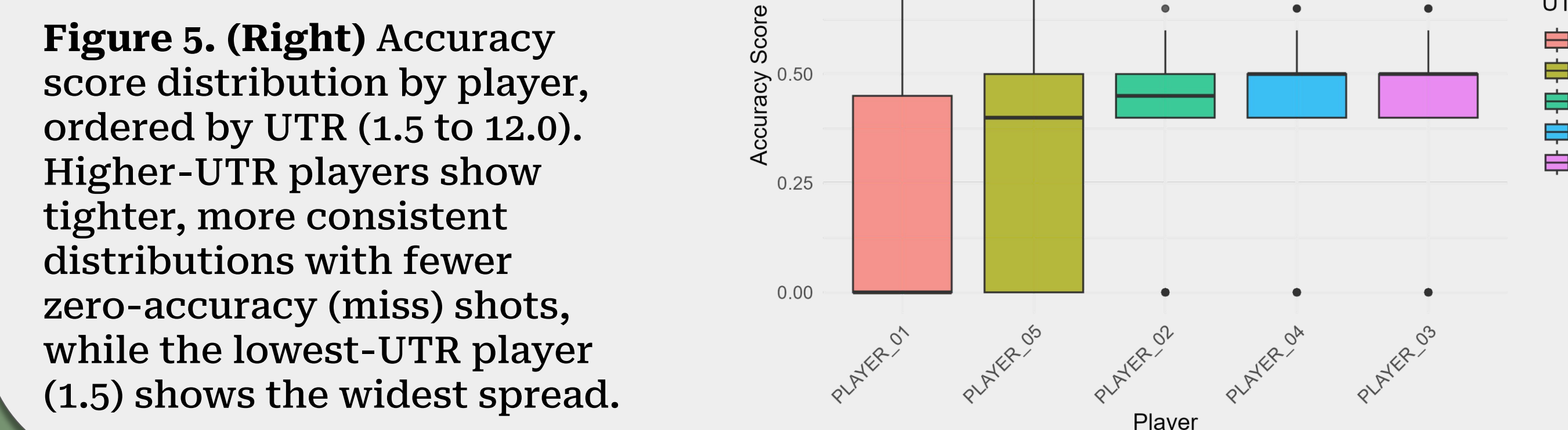


Figure 5. (Right) Accuracy score distribution by player, ordered by UTR (1.5 to 12.0). Higher-UTR players show tighter, more consistent distributions with fewer zero-accuracy (miss) shots, while the lowest-UTR player (1.5) shows the widest spread.

Discussion and Conclutions

Discussion: Stroke type and fatigue each showed small relationships with accuracy, but the blocked data collection protocol created severe multicollinearity ($r = -0.94$), preventing their independent effects from being separated. As a result, predictive models did not outperform a simple baseline, while player skill (UTR) and shot direction emerged as the strongest correlates of accuracy.

Limitations & future work: Exact measurements of ball placement has potential human error / Fatigue Proxy Confounding. Using SwingVision AI to get coordinates can be inaccurate, due to the sample size; the quirks and techniques of participants is much more impactful on the data, Expanding data variables, observing accuracy during matches and somehow applying an experiment to match play, play style archetypes

Conclusions: Skill (UTR) and shot direction, not stroke type or fatigue, drove accuracy in this dataset (UTR = 66-72% of feature importance vs. ~0% for stroke type). A blocked collection design confounded stroke type and fatigue almost completely ($r = -0.94$); future work should interleave stroke order to finally isolate their true effects.

References and Repository

<https://github.com/lcasano26/data510-Luca-Capstone>
Key dependencies: Python (pandas, scikit-learn, matplotlib), R (tidyverse, corplot, car)
References: [1] G. Fiorilli et al., Effects of a tennis match on perceived fatigue, jump and sprint performances on recreational players, The Open Sports Sciences Journal, 2020. [2] D. Whiteside & M. Reid, A shot taxonomy in the era of tracking data in professional tennis, Journal of Sports Sciences, 2018.
[3] Y. Lei et al., Rhythms of victory: Predicting professional tennis matches using machine learning, IEEE Access, 2024. [4] P. O'Donoghue, The most important points in grand slam singles tennis, Research Quarterly for Exercise and Sport, 2001.